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Report on the Research Internship: Investigating RAFT-Polymerization and PISA on Poly(Lipoates)

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1 Introduction

1.1 Monomer

1.2 RAFT

1.3 PISA

2 Experimental

2.1 Methods

Within this work a variety of methods have been used. In chapter 1, some of the theoretical background for the reactions has been displayed, whereas the focus lies on the analytical aspects in this chapter. Different instruments have been used for monitoring the reaction course, as well as the chemical and physical properties of the monomers and polymers. The following sections give more insight on the selected methods and for some of them a description of the handling of the analysis data to obtain or display results.
For writing this report, the translation tool *DeepL Translator* has been used for formulaic purposes.[1]

2.1.1 NMR

The technique nuclear magnetic resonance (NMR) spectroscopy can be used to determine the chemical composition, and by that also the purity, of chemicals, to obtain the conversion of a and monitor the polymerization reaction.
For the analysis, ^1H-NMR measurements were performed by the CeMSA@HHU (Center for Molecular and Structural Analytics @ Heinrich Heine University) on the 600 MHz spectrometer. The results have been displayed and analyzed via the software MestReNova v17.0.1 which has been obtained through the chemistry department at Heinrich Heine University.

2.1.1.1 DOSY

Diffusion-ordered spectroscopy (DOSY) is a form of NMR spectroscopy wherein the different signals are measured as a function of their chemical shift as well as their diffusion coefficient (D). This allows the differentiation between different molecules within a mixture which makes it a powerful analysis for analyzing block copolymers. Within the CeMSA@HHU, the measurments have been performed on the 600 MHz Evo Neo spectrometer. For the analysis, the software TopSpin 5.0.0 Processing by Bruker has been used. The software can be acquired through the academic processing license on the Bruker website.[2]
The analysis of DOSY spectra can be done the following way:

1. Make a copy of the folder with the raw data
2. Navigate to "*/pdata/1*" within the copied raw data folder and localize the file called "*2rr*"
3. Open TopSpin
4. Drag and drop the file "*2rr*" into TopSpin

5. Enter the following commands into the command line after the previous one is done:
    a) xf2
    b) abs2
    c) eddosy
    d) setdiffparm
    e) dosy2d
6. TopSpin can be closed once the commands have been run and MestReNova can be opened
7. The edited folder with the data can be dragged and dropped into MestReNova
8. In the *Stacked* tab, the button *DOSY Plot* needs to be pressed
9. The display of the spectra can be edited via the *Properties* window as usual

2.1.2 SEC

2.1.3 MALDI-ToF-MS

2.1.4 DLS

3 Results & Discussion

4 Summary & Outlook

Acronyms

D	diffusion coefficient
DOSY	diffusion-ordered spectroscopy
NMR	nuclear magnetic resonance

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Bibliography

[1] DeepL Translator, `https://www.deepl.com/en/translator` (visited on 26/10/2026).

[2] TopSpin | NMR Data Analysis, `https://www.bruker.com/en/products-and-solutions/mr/nmr-software/topspin.html` (visited on 26/10/2026).

Appendix

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